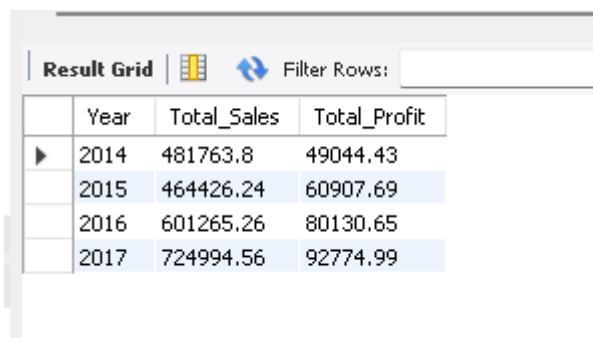


SUPER-STORE ANALYSIS

**Question-1. Calculate the year-over-year (YoY) growth in sales and profit
Display a year-wise comparison.**

Queries :

```
SELECT c.Year, ROUND(SUM(s.Sales),2) AS Total_Sales,  
        ROUND(SUM(s.Profit),2) AS Total_Profit  
FROM Sales AS s  
JOIN Calender_file AS c  
ON s.Order_Date = c.Date  
GROUP BY c.Year  
ORDER BY Year;
```



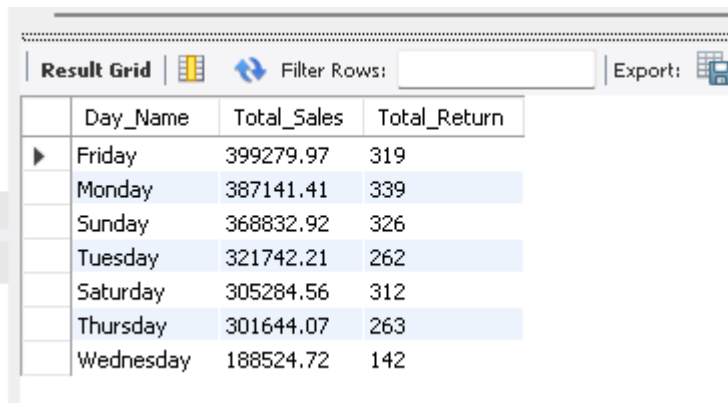
The screenshot shows a 'Result Grid' window with a table of data. The table has four columns: 'Year', 'Total_Sales', and 'Total_Profit'. There are four rows of data for the years 2014, 2015, 2016, and 2017. The values show a consistent year-over-year increase in both sales and profit.

	Year	Total_Sales	Total_Profit
▶	2014	481763.8	49044.43
	2015	464426.24	60907.69
	2016	601265.26	80130.65
	2017	724994.56	92774.99

Question-2. Identify which day of the week records the highest sales and which day has the highest number of product returns.

Queries :

```
SELECT c.Day_Name, ROUND(SUM(sales),2) AS Total_Sales,  
       SUM( CASE WHEN s.Returned = "YES" THEN 1 ELSE 0 END) AS Total_Return  
FROM sales AS s  
JOIN calender_file AS c  
ON s.Order_Date = c.Date  
GROUP BY c.Day_Name  
ORDER BY Total_Sales DESC;
```



The screenshot shows a database interface with a 'Result Grid' tab. It displays a table with three columns: 'Day_Name', 'Total_Sales', and 'Total_Return'. The data is sorted by 'Total_Sales' in descending order. The days listed are Friday, Monday, Sunday, Tuesday, Saturday, Thursday, and Wednesday. Friday has the highest sales and returns, while Wednesday has the lowest.

Day_Name	Total_Sales	Total_Return
Friday	399279.97	319
Monday	387141.41	339
Sunday	368832.92	326
Tuesday	321742.21	262
Saturday	305284.56	312
Thursday	301644.07	263
Wednesday	188524.72	142

Question-3. Determine the best-performing quarter (e.g., 2017 Q4) based on the highest total sales.

Queries :

```
SELECT c.Year,  
       c.`Quarter(Q)`,  
       ROUND(SUM(s.sales),2) AS Total_Sales  
FROM sales AS s  
JOIN calender_file AS c  
ON c.Date = s.Order_Date  
GROUP BY c.Year, c.`Quarter(Q)`  
ORDER BY Total_Sales DESC  
LIMIT 1;
```

Result Grid			
	Year	Quarter(Q)	Total_Sales
▶	2017	Q4	209130.94

Question-4. Identify the categories and sub-categories where excessive discounts resulted in negative profit.

Queries :

```
SELECT Category, `Sub-Category`,
        ROUND(SUM(Sales),2) AS Total_Sales,
        ROUND(AVG(Discount)*100,2) AS Average_Discount_PCT,
        ROUND(SUM(Profit),2) AS Total_Profit
FROM sales
GROUP BY Category, `Sub-Category`
ORDER BY Total_Profit;
```

Result Grid					
	Category	Sub-Category	Total_Sales	Average_Discount_PCT	Total_Profit
▶	Furniture	Tables	206965.53	26.13	-17725.48
	Furniture	Bookcases	114880	21.11	-3472.56
	Office Supplies	Supplies	45952.47	7.61	-1348.57
	Office Supplies	Fasteners	3008.66	8.3	942.44
	Technology	Machines	189238.63	30.61	3384.76
	Office Supplies	Labels	12486.31	6.87	5546.25
	Office Supplies	Envelopes	15339.49	8.13	6460.87
	Office Supplies	Art	27118.79	7.49	6527.79
	Furniture	Furnishings	82752.23	13.76	11588.64
	Office Supplies	Appliances	107532.16	16.65	18138.01

Question-5. Calculate the return rate (%) for each region and identify the region with the highest product return rate.

Queries :

```
SELECT Region, COUNT(*) AS Total_Orders,
       SUM(CASE WHEN Returned = "Yes" THEN 1 ELSE 0 END) AS Return_Orders,
       ROUND((SUM(CASE WHEN Returned = "Yes" THEN 1 ELSE 0 END) / COUNT(*)) *
       100, 2) AS Return_Rate_PTC
FROM sales
GROUP BY Region
ORDER BY Return_Rate_PTC DESC;
```

Question-6. Identify the top 10 customers who generated the highest overall losses for the company.

Queries :

```
SELECT Customer_ID, Customer_Name,
       ROUND(SUM(Sales),2) AS Total_Sales,
       ROUND(SUM(Profit),2) AS Total_Loss
FROM sales
GROUP BY Customer_ID, Customer_Name
HAVING Total_Loss<0
ORDER BY Total_Loss
LIMIT 10;
```

Result Grid				
Filter Rows:				
Export: 				
	Customer_ID	Customer_Name	Total_Sales	Total_Loss
▶	CS-12505	Cindy Stewart	5690.05	-6626.39
	GT-14635	Grant Thornton	9351.21	-4108.66
	LF-17185	Luke Foster	3930.51	-3583.98
	SR-20425	Sharelle Roach	3233.48	-3333.91
	HG-14965	Henry Goldwyn	3184.17	-2798.59
	NC-18415	Nathan Cano	2218.99	-2204.81
	SB-20290	Sean Braxton	8057.89	-2082.75
	SM-20320	Sean Miller	25043.05	-1980.74
	CP-12340	Christine Phan	5888.28	-1850.3
	BM-11140	Becky Martin	11609.9	-1711.43

Question-7. Shipping Delay Analysis (Calculate the average delivery time (In days) for each shipping mode.)

Queries :

```
SELECT Ship_Mode,
       ROUND(AVG(DATEDIFF(Ship_Date, Order_Date)),2) AS AVG_Shipping_Day
FROM sales
GROUP BY Ship_Mode
ORDER BY AVG_Shipping_Day;
```

Result Grid		
Filter Rows:		
Export: 		
	Ship_Mode	AVG_Shipping_Day
▶	Same Day	0.91
	First Class	23.31
	Second Class	30.49
	Standard Class	41.96

Question-8. Analyze whether late deliveries (more than 3 days) lead to a higher product return rate by comparing on-time deliveries with late deliveries.

Queries :

SELECT

CASE

WHEN DATEDIFF(Ship_Date, Order_Date) > 3 THEN 'Late'

ELSE 'On_Time' END AS Delivery_Status,

COUNT(*) AS Total_Orders,

SUM(Returned = 'Yes') AS Return_Orders,

ROUND(SUM(Returned = 'Yes') * 100.0 / COUNT(*), 2) AS Return_Rate_PCT

FROM sales

GROUP BY Delivery_Status;

Result Grid				
		Filter Rows:	Export:	Wrap Ce
	Delivery_Status	Total_Orders	Return_Orders	Return_Rate_PCT
►	Late	6094	1210	19.86
	On_Time	3600	753	20.92

Question-9. Salesperson Performance (Identify the salesperson with the highest sales and the salesperson with the lowest profit margin (%).)

Queries :

```
SELECT
    Retail_Sales_People,
    ROUND(SUM(Sales),2) AS Total_Sales,
    ROUND(SUM(Profit),2) AS Total_Profit,
    ROUND((SUM(Profit) / SUM(Sales)) * 100, 2) AS Profit_Margine_PCT
FROM sales
GROUP BY Retail_Sales_People
ORDER BY Total_Sales;
```

Result Grid   Filter Rows: Export:  Wrap C				
	Retail_Sales_People	Total_Sales	Total_Profit	Profit_Margine_PCT
▶	Cassandra Brandow	388983.59	46035.69	11.83
	Kelly Williams	497800.87	40128.9	8.06
	Chuck Magee	672194.05	90672.01	13.49
	Anna Andreadi	713471.34	106021.15	14.86

Question-10. Identify customers who placed orders in 2015 or 2016 but did not place any orders in 2017.

Queries :

```
SELECT DISTINCT s.Customer_ID, s.Customer_Name, s.Product_Name, s.Order_Date
FROM sales AS s
JOIN calender_file AS c
ON s.Order_Date = c.Date
WHERE c.Year IN (2015, 2016)
AND s.Customer_ID NOT IN (
    SELECT s.Customer_ID
    FROM sales AS s
    JOIN calender_file AS c
    ON s.Order_Date = c.Date
    WHERE c.Year = 2017
);
```

Result Grid				
Filter Rows:		Export: 		
Wrap Cell Content:				
	Customer_ID	Customer_Name	Product_Name	Order_Date
	ON-18715	Odella Nelson	Wilson Jones Leather-Like Binders with DublLock...	2016-11-03
	KD-16270	Karen Daniels	Snap-A-Way Black Print Carbonless Ruled Spee...	2016-04-06
	ER-13855	Elpida Rittenbach	Fellowes Basic Home/Office Series Surge Protec...	2016-06-09
	JD-15895	Jonathan Doherty	Howard Miller 13-3/4" Diameter Brushed Chrom...	2016-05-09
	SP-20650	Stephanie Phelps	Southworth 25% Cotton Antique Laid Paper & E...	2015-05-31
	SP-20650	Stephanie Phelps	Xerox 1883	2015-05-31
	SP-20650	Stephanie Phelps	Tenex Personal Project File with Scoop Front De...	2015-05-31
	MP-17470	Mark Packer	Atlantic Metals Mobile 4-Shelf Bookcases, Cust...	2015-12-10
	MP-17470	Mark Packer	I Need's 3d Hello Kitty Hybrid Silicone Case Cov...	2015-12-10
	MP-17470	Mark Packer	Xerox 205	2015-12-10

THANK YOU